

Airbnb Price Prediction with Machine Learning

2025-03-27

Introduction

This investigation's aim is to utilize both linear and non-linear data science methods to predict prices for Airbnb listings within London. The dataset being explored contains the last trimester of data from Inside Airbnb, collected on 11/12/2024.

A structured framework was followed in this analysis, beginning with data preprocessing consisting of addressing irrelevant variables, variable transformations, missing values, and outliers. Exploratory data analysis followed, with the purpose of gaining insight into variable distributions and relationships. Next, linear modelling with OLS and Elastic Net was conducted. The implementation of non-linear models followed, with Random Forest and Feedforward Neural Network. Finally, the methods applied were evaluated in terms of predictability and interpretability.

Data Preprocessing

Irrelevant variables were first removed, such as ID and URL fields with low potential for price prediction. Groups with clear collinearity, such as several "number_of_reviews" metrics, had correlation matrices run with only the variables most informative of price being kept. Long character variables were dropped as in-depth text analysis was not part of this investigation.

Variable transformation included turning logical types to integers. Character variables were appropriately transformed to numeric or factor types. Notably this included splitting "amenities" into a collection of integer dummies and performing occurrence range filtering to keep those in the 25% to 75% range, removing overly sparse and common amenities with low predictive power. Importantly, price had log transformation applied to handle strong positive skewness.

Next, missing values for price were dropped. Variables with high percentages of missing values were dropped entirely, as they lack predictive power. Mean and median imputation were appropriately applied to variables with predictive potential containing some degree of missing values, allowing their inclusion while staying robust to skewness. Finally, outliers were removed from continuous variables to reduce distortion.

Exploratory Data Analysis (EDA)

EDA began with feature selection organized by variable type: continuous, binary, and categorical.

Continuous variables were ranked based on the product of their variance and correlation with log price, with the notion that the variance carries information and the correlation captures a strong relation to price (Figure 1). A correlation threshold of 0.05 was implemented to filter out uninformative variables.

Binary features were ordered by their mean difference in log price combined with t-tests for significance. The 10 with the greatest significant mean difference were chosen (Figure 2). Categorical variables were similarly chosen based on the spread in log price between categories (Figure 3).

These three groups combined for 22 variables post feature selection, which were explored further in EDA.

Figure 1 - Continuous Feature Selection

variable	variance	corr_with_ln_price	score
host_listings_count	2816.3546315	0.1150265	323.9553446
accommodates	4.3613268	0.6238093	2.7206362
beds	1.8378960	0.4637088	0.8522485
bedrooms	0.9731057	0.4929216	0.4796648
reviews_per_month	1.9588356	-0.1182843	0.2316995
bathrooms	0.4402429	0.3363134	0.1480596
review_scores_location	0.1179165	0.1227280	0.0144717
review_scores_cleanliness	0.1800086	0.0513357	0.0092409
longitude	0.0110910	-0.0994291	0.0011028

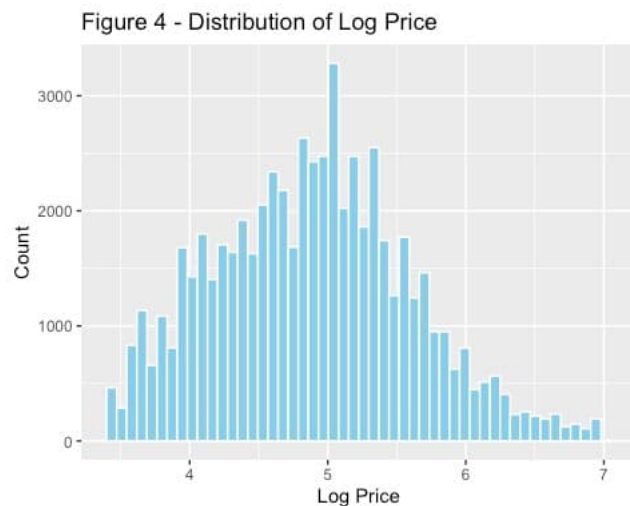
Figure 2 - Binary Feature Selection

variable	mean_diff	p_value
has_Dishwasher	0.4954811	0
has_TV	0.3314144	0
has_Hair_dryer	0.2969180	0
has_Private_entrance	0.2920543	0
has_Dining_table	0.2836068	0
has_Wine_glasses	0.2789261	0
has_Freezer	0.2664949	0
has_Iron	0.2579479	0
has_Oven	0.2441844	0
has___Cooking_basics_	0.2407597	0

Figure 3 - Categorical Feature Selection

variable	group_ln_price_sd	n_levels
room_type	0.5634381	4
property_type_1	0.4894295	18
neighbourhood_cleansed	0.2810682	33

Numeric variable distributions were inspected and treated, such as transforming to log form if positively skewed (Figure 4, see Appendix A for detail).



Insight into the relationship between features and log price was gained through scatter plots or boxplots depending on variable type (Figures 5-7, see Appendices B-D for detail).

Figure 5 - Scatterplot of Reviews per Month by Log Price



Figure 6 - Boxplot of Has Dishwasher by Log Price

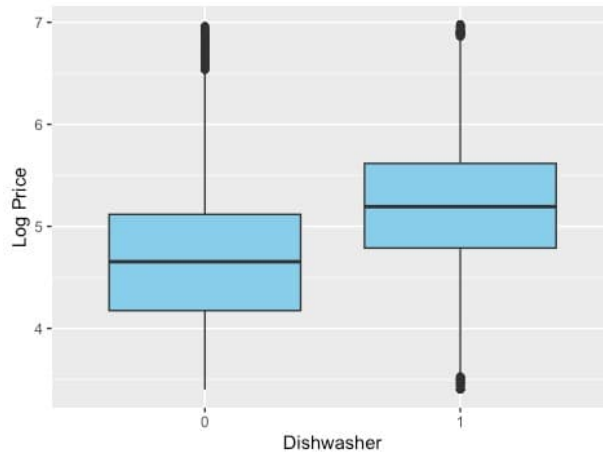
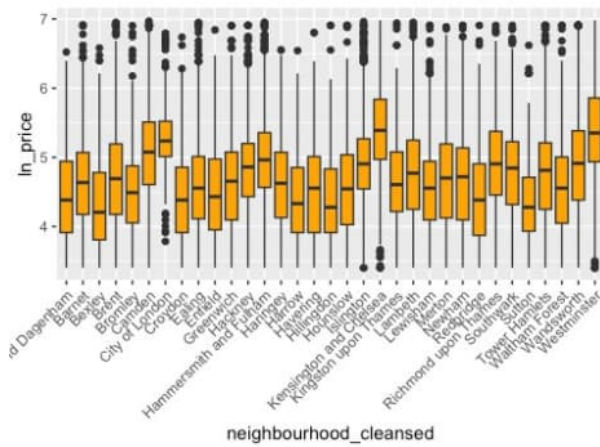


Figure 7 - Boxplot of Neighbourhood by Log Price



Next, a correlation matrix was formed to inspect for multicollinearity between continuous features (Figure 8). Finding some high correlations, Principal Component Analysis was carried out to reduce dimensionality. The resulting scree plot found that over 90% of cumulative variance was explained by the first 6 principal components (Figure 9).

Figure 8 - Correlation Matrix of Continuous Features

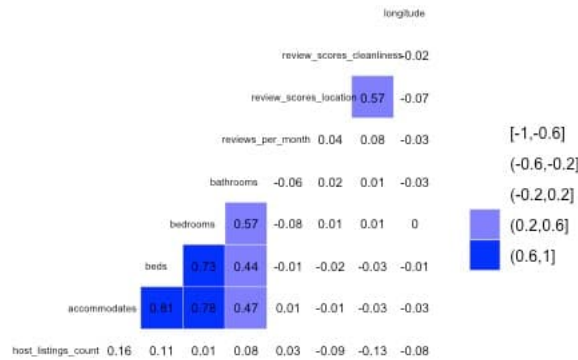
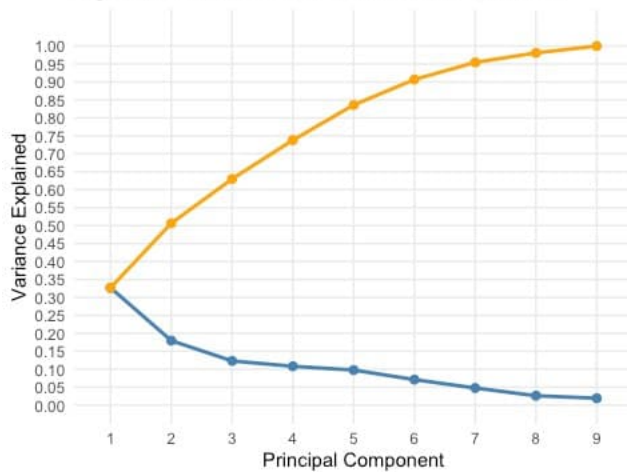


Figure 9 - Scree Plot with Cumulative Variance



Before modelling, the dataset was split 80/20 into a training set, for model fitting and parameter tuning, and a test set, for evaluating model performance.

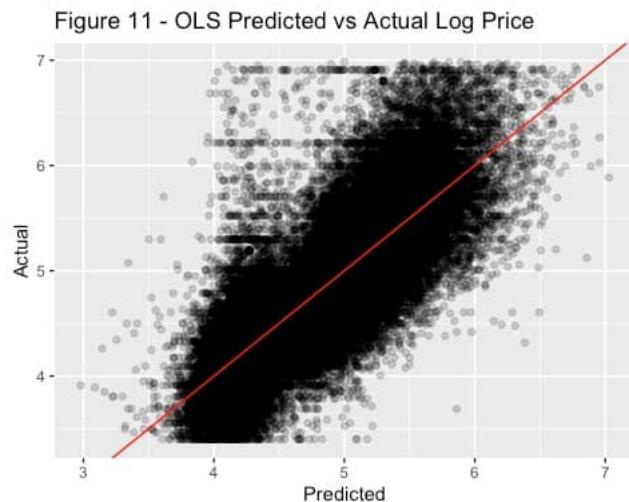
Linear Models

The first linear model applied was OLS estimation, which functions by minimizing the residual sum of squares. Two OLS models were run on log price, one with the first 6 principal components of PCA and the other with standardized features. OLS with standardized variables had a greater R-squared value (Figure 10), meaning more predictive power. It also produced more interpretable results compared to the black-box PCA approach. Thus, the standardized feature approach was used in subsequent models.

The solid fit of the OLS model with standardized variables is seen in the log price predictions scatterplot (Figure 11), with an alignment of points forming a clear linear pattern around the 45-degree line.

Figure 10 - OLS R-Squared Comparison

Model	R_squared
PCA	0.4050
Standardized Variables	0.6196



The application of Elastic Net followed to attempt to improve upon OLS performance. By combining the absolute value penalty of LASSO with the squared penalty of Ridge, Elastic Net is well suited to balance variable selection and coefficient shrinkage. This combined regularization allows it to handle correlated predictors more effectively than OLS, enabling the inclusion of slightly more variables. Notably, the high cardinality categorical variable “neighbourhood_cleansed” was retained in this model. Penalty parameter tuning was conducted with five-fold cross validation.

The Elastic Net log price predictions scatterplot (Figure 12) suggests improved performance compared to OLS with points appearing more tightly clustered around the 45-degree line. In addition to relatively strong predictive power, Elastic Net is easily interpretable, retaining and ranking individual features by importance, thus offering clear insight into key drivers of log price (Figure 13).

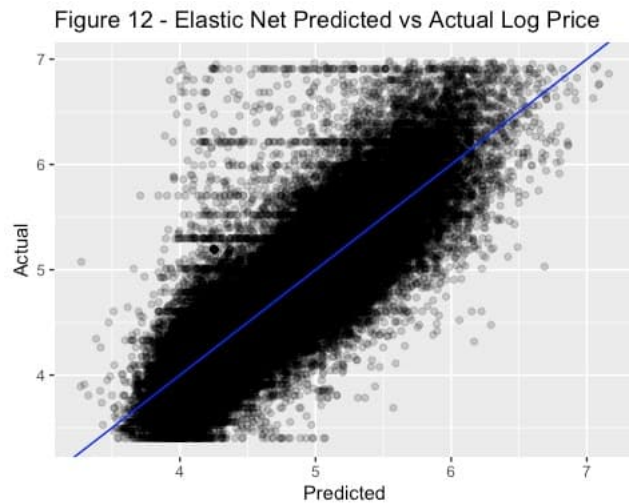


Figure 13 - Top 10 Most Predictive Features by Elastic Net

Feature	Magnitude
neighbourhood_cleansed_Westminster	0.2803
neighbourhood_cleansed_Kensington.and.Chelsea	0.2200
accommodates	0.2086
room_type_Private.room	0.1734
neighbourhood_cleansed_Camden	0.1588
neighbourhood_cleansed_Hammersmith.and.Fulham	0.1075
neighbourhood_cleansed_Wandsworth	0.1039
neighbourhood_cleansed_Islington	0.1021
bedrooms	0.1007
reviews_per_month	0.0957

When comparing the goodness of fit of linear models, Elastic Net clearly stands out as the stronger of the two. With lower RMSE and MAE, and an R-squared value that is 0.0753 higher, it performs better than OLS in every metric (Figure 14).

Figure 14 - Linear Models Goodness of Fit

model	rmse	rsq	mae
Elastic Net	0.3960	0.6913	0.2963
OLS	0.4396	0.6196	0.3342

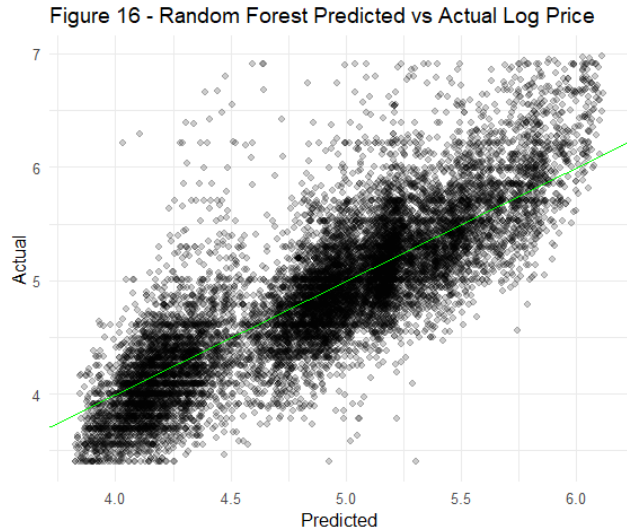
Non-Linear Models

Next, non-linear models were applied to capture more complex relationships between features and log price, starting with Random Forest. This method builds multiple decision trees through bootstrap aggregation and random feature selection, averaging their predictions for improved accuracy and reduced risk of overfitting. Random Forest can uniquely capture non-linear feature interactions without requiring explicit transformation, allowing the use of all variables in the training set. The best performing Random Forest had 300 trees, choosing 16 predictors at each split. To reduce the risk of overfitting, minimum node size was set to 5 and maximum nodes was set to 80.

Random Forest has high interpretability as it can rank feature importance by increase in MSE percentage, indicating how much prediction error increases when the given feature is permuted (Figure 15). Visually, its prediction scatterplot (Figure 16) indicates strong fit, with a high clustering of points around the 45-degree line, appearing to be more accurate than the linear models. This is supported by a 0.7154 R-squared, the highest of any model so far.

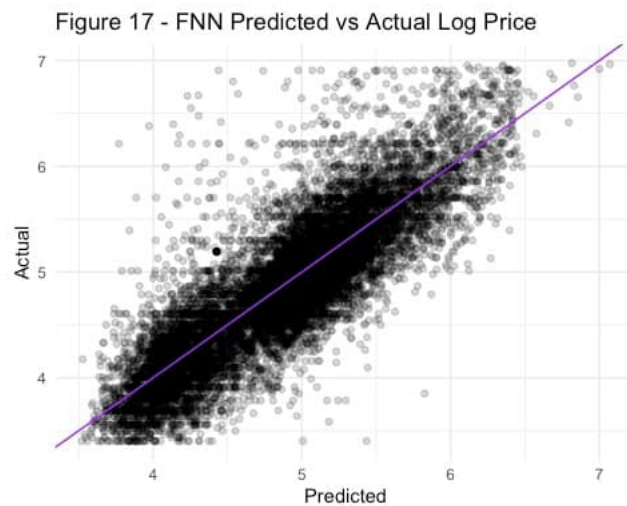
Figure 15 - Top 10 Most Predictive Features by Random Forest (%IncMSE)

Feature	Importance (%IncMSE)
neighbourhood_cleansed	44.97
bathrooms	28.75
property_type_1	28.19
availability_90	25.00
review_scores_location	24.69
host_response_time	23.53
accommodates	22.76
bedrooms	21.48
room_type	20.96
latitude	20.39



The final model applied was a Feedforward Neural Network (FNN). FNNs are composed of layers of interconnected nodes with non-linear activation functions, in this case ReLU, enabling the model to learn complex patterns. Standardized features were used to train a single hidden layer, due to computational constraints, with 5 units, determined through iteration. Weight decay regularization helped reduce risk of overfitting.

The FNN prediction scatterplot (Figure 17) shows the tightest clustering around the 45-degree line from all models, indicating the best fit and predictive performance, which is backed up by the highest R-squared value of all models at 0.7179.



FNN outperformed Random Forest in all goodness of fit tests, with higher R-squared and lower RMSE and MAE (Figure 18). However, Random Forest still holds great validity from its interpretability, which the black-box FNN lacks.

Figure 18 - Non-Linear Models Goodness of Fit

model	rmse	rsq	mae
Random Forest	0.3806	0.7154	0.2876
FNN	0.3746	0.7179	0.2801

Evaluation

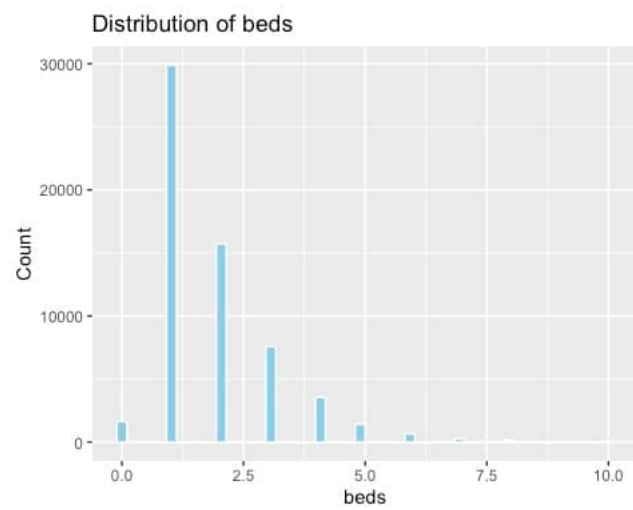
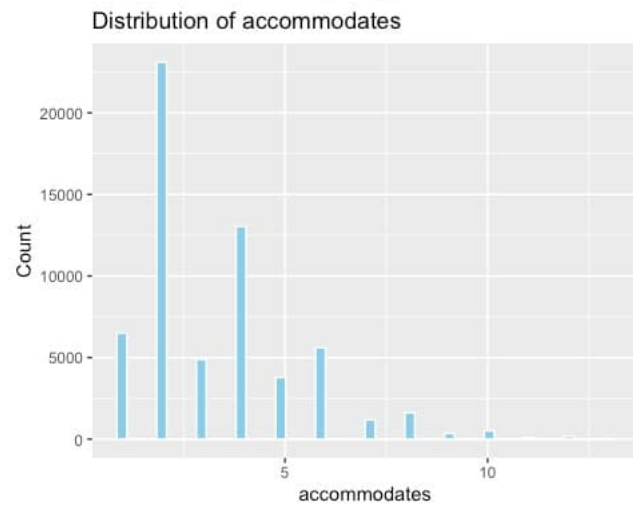
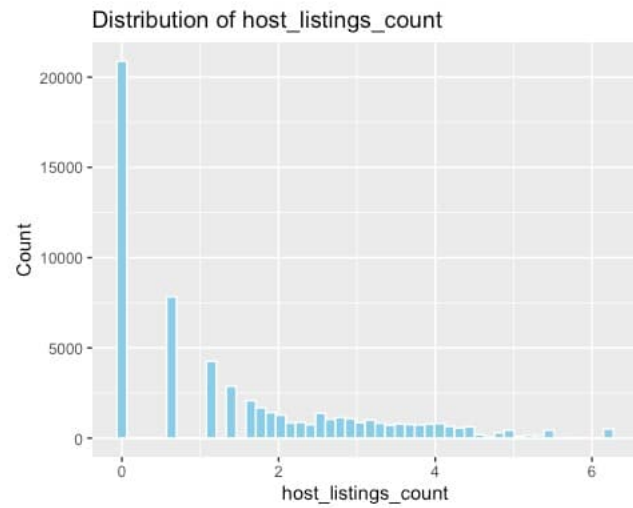
Model performance consistently improved as complexity increased, with OLS being the weakest and FNN being the strongest across all fit metrics (Figure 19). While FNN dominated prediction strength, Random Forest and Elastic Net offered great interpretability. Both identified neighborhoods, accommodates, and bedrooms as key features of price (Figures 13 and 15).

Figure 19 - Comparison of Goodness of Fit

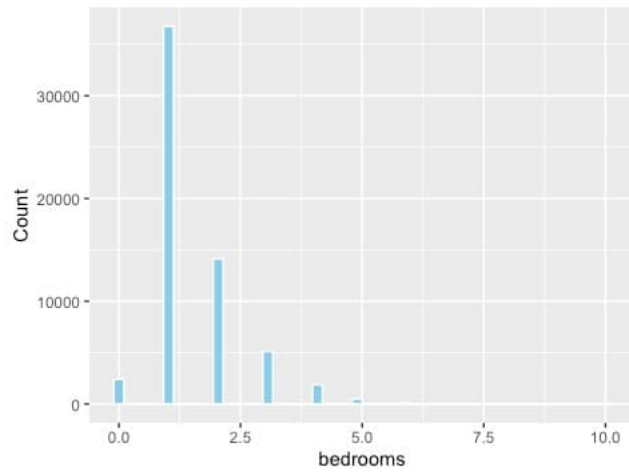
model	rmse	rsq	mae
FNN	0.3746	0.7179	0.2801
Random Forest	0.3806	0.7154	0.2876
Elastic Net	0.3960	0.6913	0.2963
OLS	0.4396	0.6196	0.3342

This investigation into predicting Airbnb listing prices in London achieved strong model performance and valuable insight into the most influential features through linear and non-linear methods. In the future, text analysis through natural language processing on listing descriptions and deeper neural network architectures could further improve robustness.

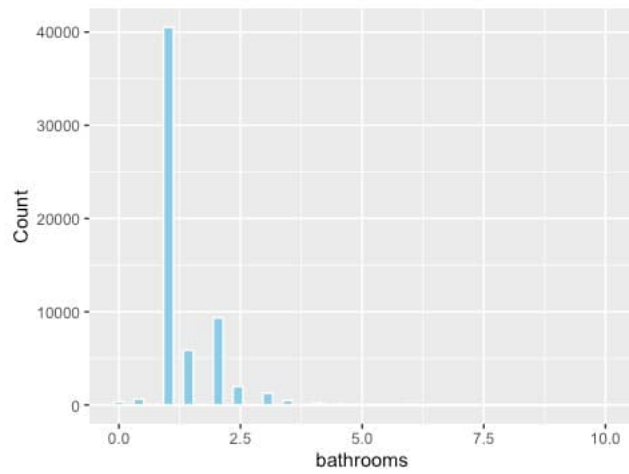
Appendix A: Distribution of Variables



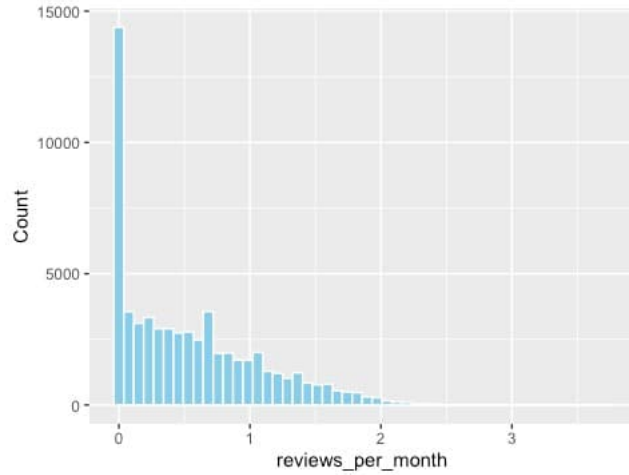
Distribution of bedrooms



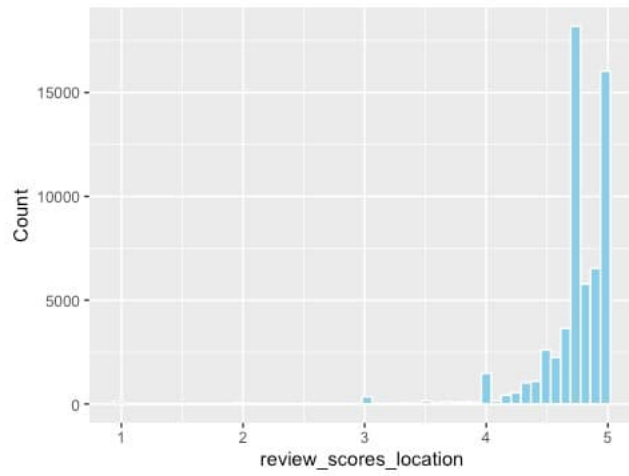
Distribution of bathrooms



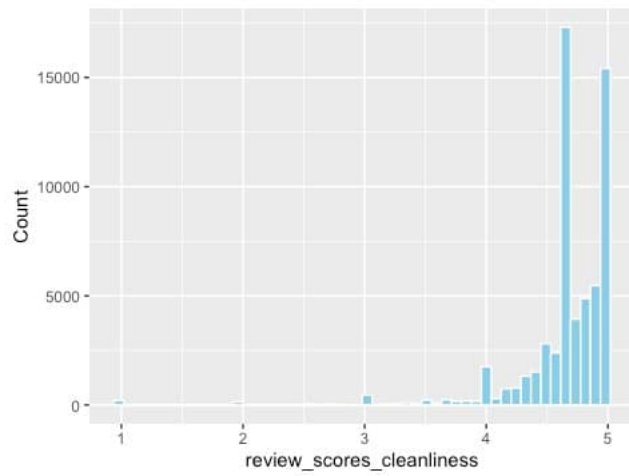
Distribution of reviews_per_month



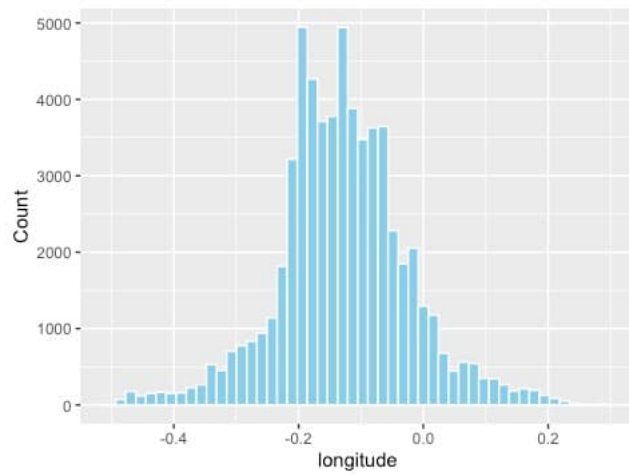
Distribution of review_scores_location



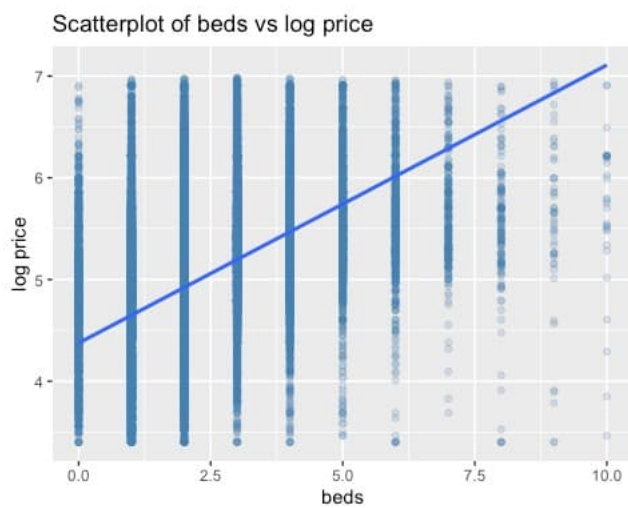
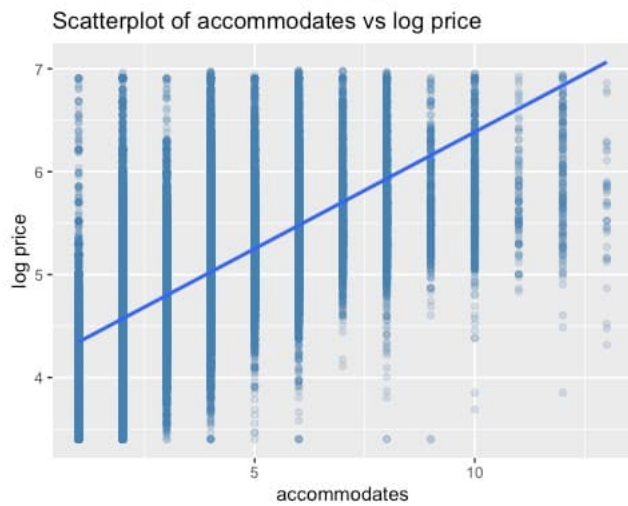
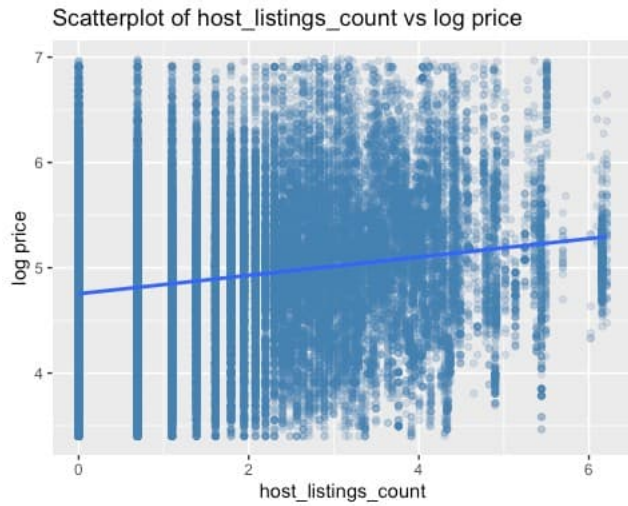
Distribution of review_scores_cleanliness



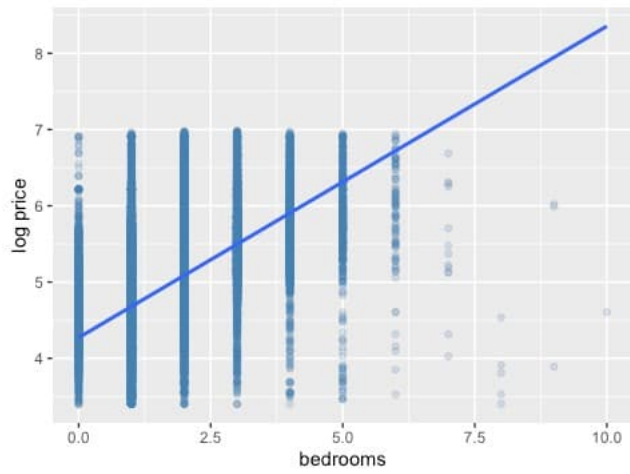
Distribution of longitude



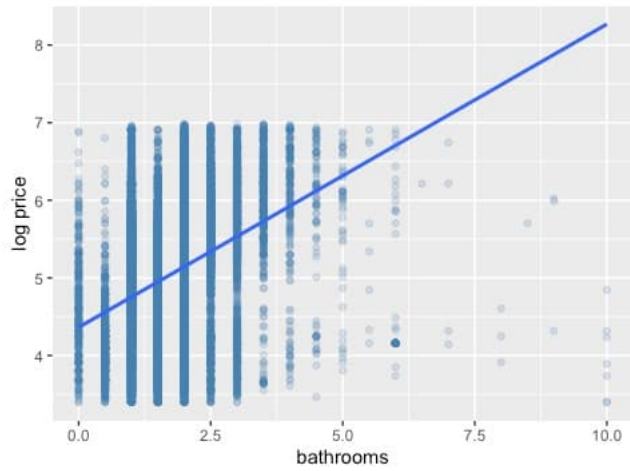
Appendix B: Scatter Plots of Continuous Features



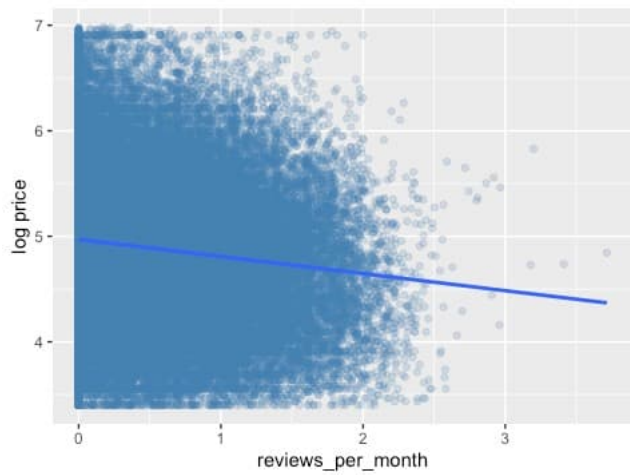
Scatterplot of bedrooms vs log price



Scatterplot of bathrooms vs log price



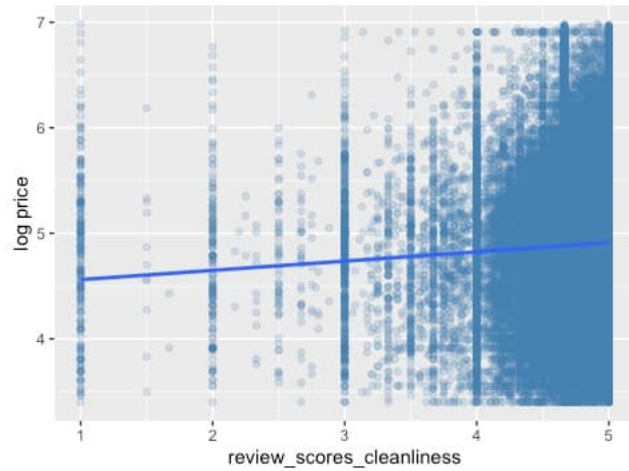
Scatterplot of reviews_per_month vs log price



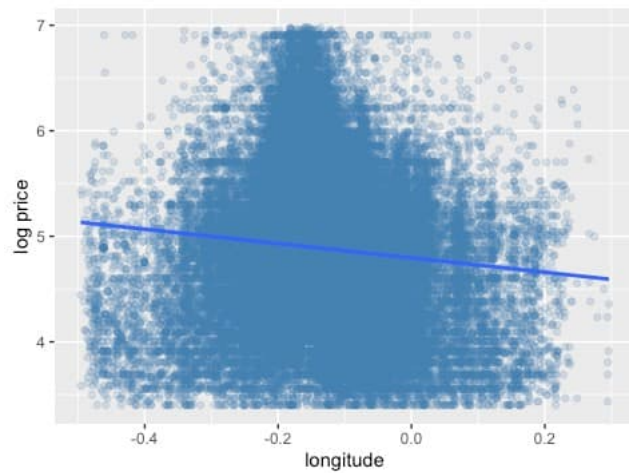
Scatterplot of review_scores_location vs log price



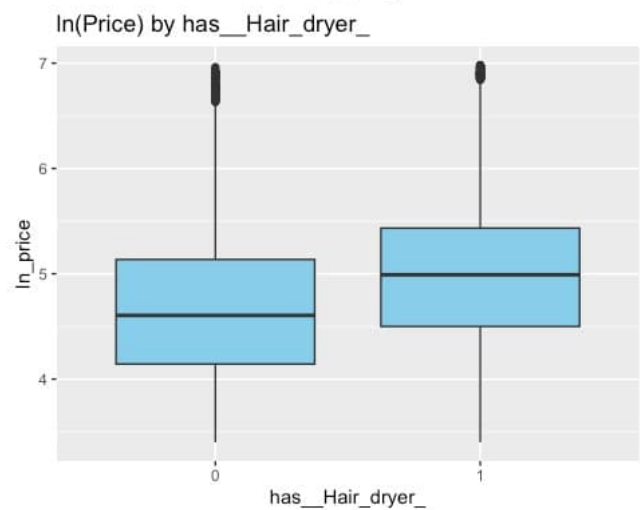
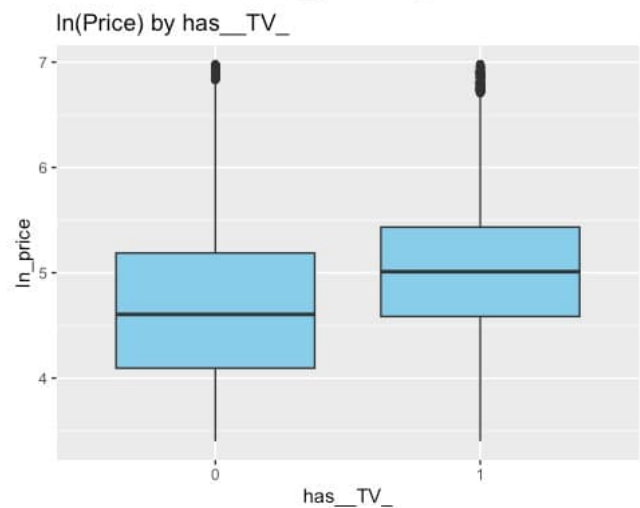
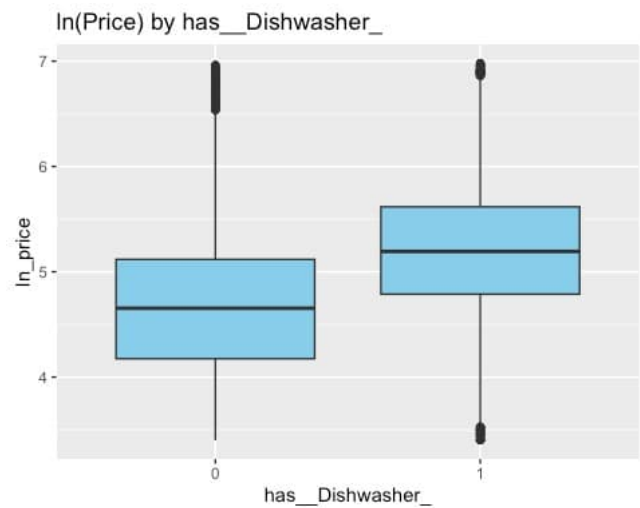
Scatterplot of review_scores_cleanliness vs log price

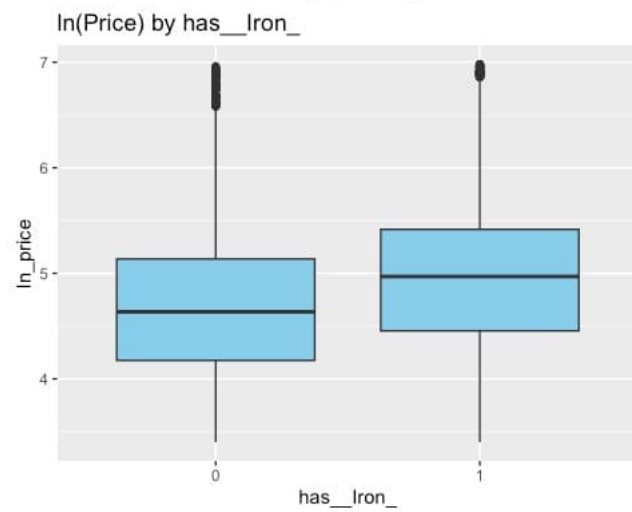
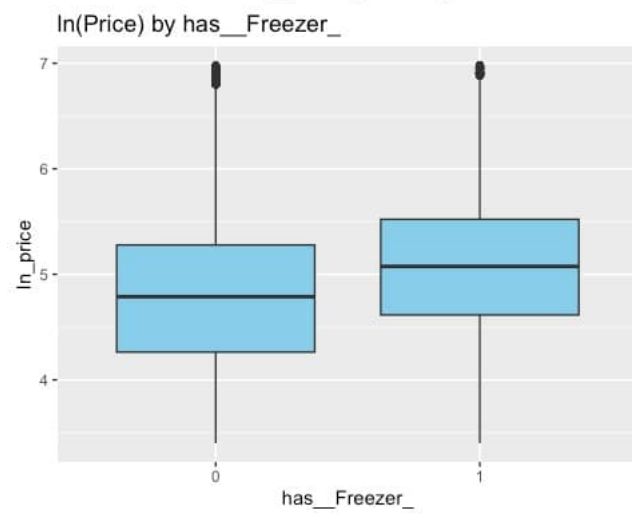
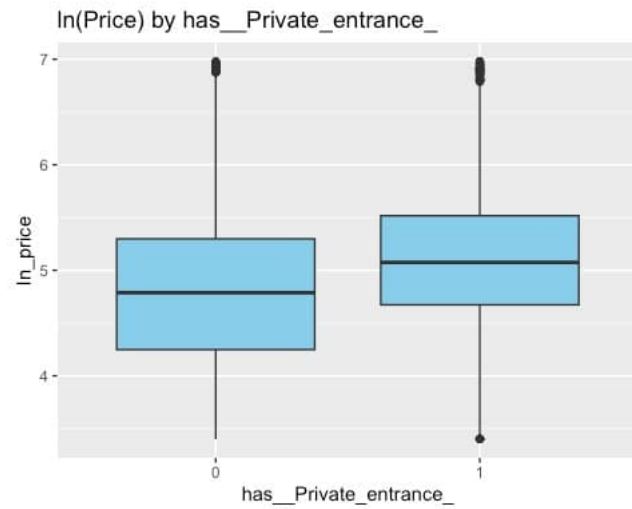


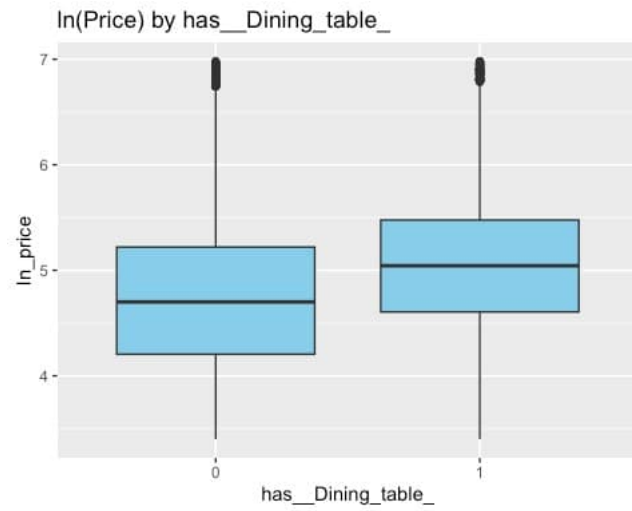
Scatterplot of longitude vs log price

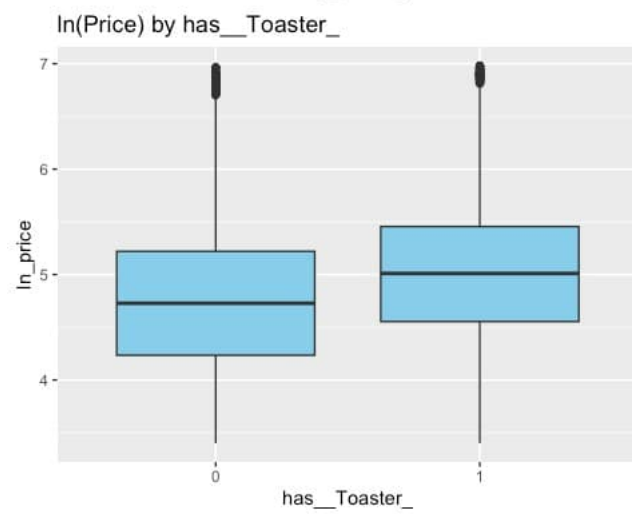
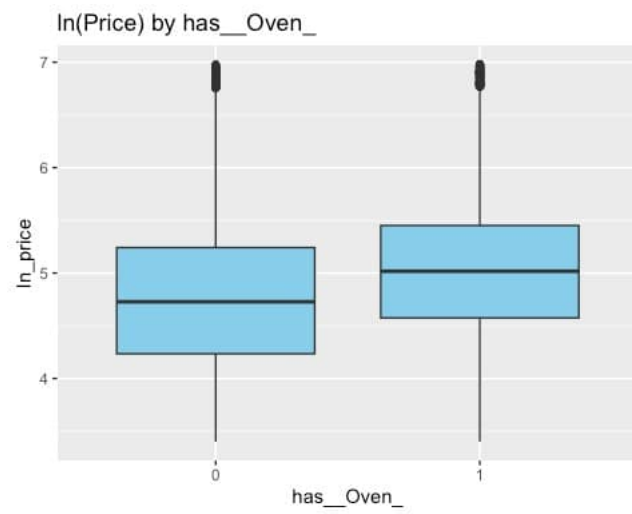


Appendix C: Binary Feature Box Plots









Appendix D: Categorical Feature Box Plots

